NASA SPACE APPS CHALLENGE CAIRO 2026 • OFFICIAL PROJECT DESCRIPTION

ECHOES OF SILENCE

Team Name: Spaghetti Coders (Omar, Bashar, Youssef) | **Selected Challenge:** Abandoned but Not Forgotten: Storytelling about NASA's Discarded Equipment on the Moon and Mars

1. PROJECT IDENTITY, LINKS & SHORT SUMMARY

Project Name: Echoes of Silence
Team Name: Spaghetti Coders
Team Members: Omar, Bashar, Youssef
Target Audience: Students (Ages 10–14), Educators & Museum Visitors

GitHub Repository (Source Code & Setup): https://github.com/omarsh4114/NASA-Abandoned-Not-Forgotten-404
Selected Challenge: Abandoned but Not Forgotten

What We Built & Problem Solved: Decades of space exploration have left dozens of inactive NASA landers, rovers, and probes resting across our Solar System. To students aged 10–14, these silent machines can seem like forgotten "space junk" rather than historic scientific monuments. **Echoes of Silence** is an interactive 3D web exhibition built with Three.js and React Three Fiber that brings 13 real NASA spacecraft across the Moon, Mars, Asteroid 433 Eros, Dwarf Planet Ceres, and Deep Space into a living museum.

How It Answers the Challenge & Why It Matters: Visitors trace real interplanetary transfer paths in the **Solar System Explorer**, compare all 13 spacecraft side-by-side on the **Orbital Museum** deck, and step onto the **1:1 Surface Stage** to inspect hardware at true metric scale (`1 unit = 1 meter`) next to a `1.75m` astronaut. Inside each exhibit, students read plain-English first-person stories, click interactive **"Ask the Spacecraft" Q&A chips**, toggle between historic surface photos and modern LROC/HiRISE satellite reconnaissance, and earn a printable **Explorer Passport Certificate**.

2. PROJECT DETAILS — HOW IT WORKS STEP-BY-STEP

Main User Task Flow (What Goes In → What the Project Does → What Comes Out):

1. **Select an Exploration Mode (Input):** From the welcome screen or top navigation bar, the visitor selects one of three 3D views: *Solar System Explorer*, *Orbital Museum*, or *1:1 Surface Stage*, and clicks any of the 13 NASA spacecraft from the bottom dock or 3D planetary pins.
2. **Interactive 3D Inspection & Telemetry Processing (What the App Does):** The camera smoothly frames the chosen spacecraft. In *Solar System Mode*, a 3D transfer probe travels along the mission trajectory curve while displaying one-way radio light-delay telemetry (e.g., `1.28 Light-Seconds` for the Moon, `~14.2 Light-Minutes` for Mars, `~23.1 Light-Minutes` for Ceres). In *1:1 Surface Stage*, the app renders the spacecraft at true metric dimensions alongside a `1.75m` astronaut and numbered 3D engineering hotspot pins.
3. **Storytelling, Q&A & Learning Outcome (Output):** Opening the *More Info* drawer reveals a 3-part first-person story, a *Mission Photo vs. Orbital Recon* image toggle, 3 clickable "*Ask the Spacecraft*" student Q&A pills with verified NASA telemetry tags, and a progress tracker that unlocks a printable *Certificate of Space Exploration* once exhibits are visited.

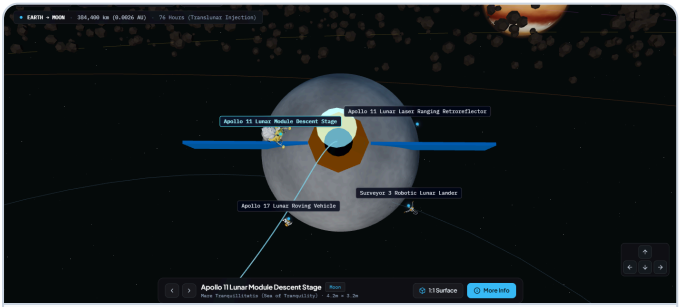


Figure 1 — Solar System Explorer: 3D heliocentric map showing planetary orbits, transfer trajectory arcs, and planetary landing site pins.

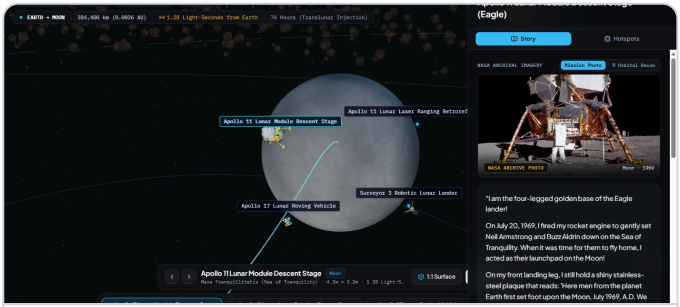


Figure 2 — Story & Interactive Q&A Drawer: Displays archival NASA imagery, radio light-delay (1.28 Light-Seconds), and student Q&A chips.

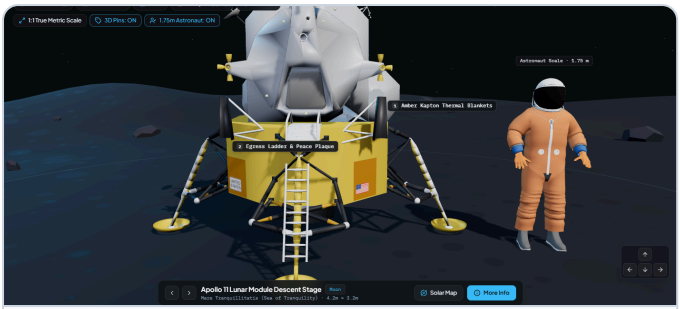


Figure 3 — 1:1 Surface Stage (Moon): Apollo 11 Eagle Descent Stage at true 1:1 metric scale beside a 1.75m astronaut reference and 3D pins.

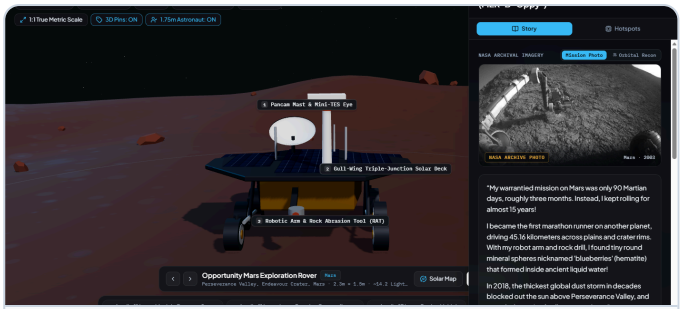


Figure 4 — 1:1 Surface Stage (Mars): Opportunity Rover (MER-B) on Martian terrain with engineering subsystem callouts.

Tools, Coding Languages & Software Used

Languages & Frameworks: TypeScript, React 19, HTML5 Canvas, CSS3 / Tailwind CSS v4, GLSL Shaders.
3D Engine & Spatial Libraries: Three.js (`r183`), React Three Fiber (`@react-three/fiber`), `@react-three/drei`, Draco 3D Mesh Decompression, WebRTC Camera API (for 1:1 AR camera passthrough), and Vite 8 build system.

Honest Separation: Working Features, Current Limits & Future Plans

What Works Now (Verified Prototype):

- All 3 interactive 3D modes (Solar System, Orbital Museum, 1:1 Surface Stage).
- All 13 NASA spacecraft with 39 3D engineering pins and 39 interactive Q&A answers.
- 1.75m Astronaut scale toggle, keyboard/touch navigation, and printable Explorer Passport.

Current Limits:

- Planetary positions in the 3D Solar System view use fixed artistic orbital anchors for clear visual framing rather than live Julian-date ephemerides.
- 6 spacecraft use optimized NASA `.glb` meshes while 7 use custom procedural multi-part 3D assemblies to keep initial page load fast.

Future Plans:

- Connect NASA JPL Horizons API for live real-time planetary positions.
- Add Arabic localization and classroom teacher lesson-plan worksheets.
- Expand the catalog to include Cassini, Huygens, and Voyager 1/2.

3. NASA DATA AND RESOURCES USED

Every coordinate, launch/end year, metric dimension, failure explanation, and archival photograph in **Echoes of Silence** comes directly from official NASA open-data repositories:

NASA Resource / Dataset	Official Source URL	How We Used It in Echoes of Silence
NASA Space Science Data Coordinated Archive (NSSDCA)	https://nssdc.gsfc.nasa.gov/	Verified official spacecraft IDs (e.g., `1969-059C`, `2003-032A`, `2018-042A`), launch/end dates, dry masses, and metric dimensions for all 13 exhibits.
NASA 3D Resources	https://nasa3d.arc.nasa.gov/	Sourced and normalized reference 3D spacecraft glTF/GLB geometry for Apollo Lunar Module, Ingenuity, InSight, Viking 1, Dawn, and Pioneer 10.
Lunar Reconnaissance Orbiter Camera (LROC)	https://www.lroc.asu.edu/	Used Narrow Angle Camera (NAC) orbital frames (`M175124932R`, `M168000580LR`, `M114110397R`) and lunar coordinates for Apollo 11, Apollo 17 LRV-3, and Surveyor 3.
Mars Reconnaissance Orbiter HiRISE	https://www.uahirise.org/	Used orbital reconnaissance imagery and landing coordinates for Ingenuity (`PIA26244`), Opportunity (`ESP_056972_1775`), Spirit (`PSP_003688_1650`), InSight (`PIA23000`), Phoenix (`PIA13146`), and Pathfinder/Sojourner (`PIA09105`).
NASA Planetary Data System (PDS)	https://pds.nasa.gov/	Used Lunar Geotechnical reports (`NASA SP-238`, `NASA SP-284`), InSight SEIS Marsquake totals (1,319 events), and Small Bodies Node data for 433 Eros and Ceres.
NASA Image and Video Library & Exoplanet Archive	https://images.nasa.gov/ https://exoplanetarchive.ipac.caltech.edu/	Sourced public-domain surface/mission photographs for the drawer's *Mission Photo vs. Orbital Recon* viewer and Kepler exoplanet discovery totals (2,662 confirmed planets).

4. OTHER RESOURCES AND SOURCE CREDITS

- **Three.js (MIT License):** https://threejs.org/ — WebGL 3D rendering, PBR materials, custom GLSL shaders, and procedural canvas textures.
- **React 19 & React Three Fiber / Drei (MIT License):** https://r3f.docs.pmnd.rs/ — Declarative 3D scene graph, camera controls, and 3D HTML annotations.
- **Tailwind CSS v4 & Lucide React Icons (MIT / ISC License):** https://tailwindcss.com/ & https://lucide.dev/ — Responsive HUD styling and UI iconography.
- **Google Draco 3D Compression (Apache 2.0 License):** https://github.com/google/draco — Mesh decompression for 3D `.glb` assets.

5. AI AND SI (SUPER INTELLIGENCE) USE DISCLOSURE

In full compliance with the **NASA Space Apps 2026 Judging Guidelines**, Team **Spaghetti Coders** states that **yes, AI/SI tools were used** during development to assist with implementation, while our team independently conceived the project idea, 3-tier spatial exhibition architecture, and educational structure:

- **Google Antigravity & Google Gemini (AI Coding & Copyediting Assistants):**
  - *Task & Affected Work:* Assisted with React Three Fiber / TypeScript code scaffolding, procedural planetary canvas texture/shader tuning (`src/utils/planetaryTextures.ts`, `src/components/solar-system/SolarSystemOrrery.tsx`), and age-appropriate (ages 10–14) copyediting of first-person spacecraft stories and Q&A chips (`src/data/nasaHardware.ts`, `src/data/artifacts.ts`).
- **Manus AI (Presentation Deck & Submission Structuring Assistant):**
  - *Task & Affected Work:* Used to structure and design our 7-slide NASA Space Apps Cairo 2026 presentation deck, cross-verify our 13-spacecraft NSSDCA telemetry summary, and align our submission materials with official judging guidelines.
- **No AI-Generated Images, Video, or Altered NASA Branding:**

- All archival mission photographs and orbital reconnaissance images displayed in the application are authentic, unmodified public-domain NASA photographs fetched from `images.nasa.gov`.
- No NASA logos, flags, or official mission insignia were generated, modified, or used in any AI-assisted content.